

Tyson Klingner

University of Washington

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Seattle, WA

Education

University of Washington

Ph.D. in Mathematics

2023–present

Seattle, WA

Advisors: Sándor Kovács and Giovanni Inchiostro

The University of Adelaide

Master of Philosophy in Mathematics

2021–2022

Adelaide, Australia

Advisor: David Baraglia

The University of Adelaide

Bachelor of Mathematical Sciences (Advanced)

2018–2020

Adelaide, Australia

Research Interests

Algebraic geometry, birational geometry, moduli spaces, and algebraic stacks.

Publications and Preprints

[1]. **Characterizing Contractions and Weighted Blowdowns**

Soham Ghosh and Tyson Klingner

Preprint, 2026. arXiv:2609.04023.

[2]. **Spectral data of special orthogonal Higgs bundles and Hecke modification**

Tyson Klingner

The Quarterly Journal of Mathematics **75** (2024), no. 4, 1529–1553. doi:10.1093/qmath/haae057.

[3]. **Evaluation of LLMs for Mathematical Formalization in Lean**

Tyson Klingner, Drew Bladek, Escher Crawford, Bohao Chen, Ariel Fu, Kaira Nair, Jarod Alper, Giovanni Inchiostro, and Vasily Ilin

Preprint, 2026. arXiv:2606.05632.

Teaching

Grader, Math 534

Autumn 2026

Instructor, Math 207

Lectured the course, prepared course materials, and wrote and administered the exams.

Summer 2026

Teaching Assistant, Math 126

Winter 2026

Teaching Assistant, Math 125

Autumn 2025

Lead Teaching Assistant, Math 208

Winter 2025

Teaching Assistant, Math 125	Autumn 2024
Teaching Assistant, Math 125	Summer 2024
Teaching Assistant, Math 125	Winter 2024
Teaching Assistant, Math 125	Autumn 2023

Additional teaching experience. Tutoring and marking in mathematics at the University of Adelaide and related programs, including complex analysis, vector calculus, integration and analysis, differential equations, and introductory mathematics (2021–2023).

Mentoring

Undergraduate Research Mentor, UW Math AI Lab 2026
 Led a group of undergraduate researchers studying large language models for mathematical formalization in Lean; the project resulted in the preprint *Evaluation of LLMs for Mathematical Formalization in Lean*.

Mentor, Washington Directed Reading Program Spring 2024
Exploring the Complex World. Mentee: Brinda Moudgalya.
 Mentored an undergraduate reading project introducing complex analysis, including contour integration, Cauchy’s theorems, and residue theory.

Talks

<i>Canonical Model of Surfaces</i> UW Student Algebraic Geometry Seminar	Spring 2026
<i>Surface Singularities</i> UW Student Algebraic Geometry Seminar	Winter 2026
<i>Uniformization of Stacky Curves</i> UW Student Algebraic Geometry Seminar	Autumn 2025
<i>Yau’s Proof of the Calabi Conjecture</i> UW Student Differential Geometry Seminar	Winter 2025
<i>Eigenvalues and Much More</i> 1-2-3 Seminar, University of Washington	Spring 2024
<i>Introduction to Complex Algebraic Geometry</i> UW Student Algebraic Geometry Seminar	November 2023
<i>Langlands Duality in the Hitchin System</i> Australian Postgraduate Algebra Colloquium	June 2022
<i>Spectral Curve Correspondence</i> Geometry and Topology Squared Seminar	May 2022
<i>Spectral data of L-twisted Higgs bundles and Langlands duality</i> AYAPS, Adelaide	March–April 2022

Academic Service

Organizer , Deformation Theory Seminar, University of Washington	Spring 2026
Organizer , Student Differential Geometry Seminar, University of Washington	2024–2025

Awards and Funding

Fulbright Australia Future Scholarship (Postgraduate)	2023–2028
University of Adelaide Research Scholarship , Master of Philosophy	2021–2022
Summer Research Project Scholarship , University of Adelaide	2020
JR Wilton Prize , University of Adelaide	2020
ES Barnes Prize , University of Adelaide	2020
David Murray Prize , University of Adelaide	2020
Dean's Recognition of Academic Excellence Award , University of Adelaide	2019, 2020